

Prj 2: Hybrid Neural Network and Fuzzy Logic-Based Intrusion Detection System

Computation Intelligence and its Applications in Mechatronics
Amirkabir University of Technology



In this project, students will design a hybrid Intrusion Detection System (IDS) that combines the capabilities of Deep Neural Networks (DNNs) for pattern recognition with the interpretability and adaptability of Fuzzy Logic Systems (FLSs). The objective is to construct a system that can reliably identify network intrusions while ensuring that the reasoning is comprehensible to humans. An important aspect of this initiative includes utilizing transfer learning by training the system on one established cybersecurity dataset and assessing its performance in detecting new, modern attacks found in a different yet compatible dataset. The project involves building and integrating both elements and evaluating their overall effectiveness.

Project Structure

1. Preprocess and ensure compatibility between the selected datasets (CIC-IDS2017 and CSE-CIC-IDS2018).
2. Train a deep neural network classifier on the primary training dataset (CIC-IDS2017) to learn to distinguish between normal and various known malicious traffic patterns.
3. Adapt and fine-tune the trained DNN using the CSE-CIC-IDS2018 dataset to detect newer attacks.
4. Develop an interpretable FLS based on selected input features or information derived from the DNN outputs.
5. Integrate the trained deep learning model and the designed FLS into a unified hybrid IDS architecture.

Dataset Description

For this project, you will work with two prominent benchmark datasets in network intrusion detection: **CIC-IDS2017** and **CSE-CIC-IDS2018**.

- **Dataset 1: CIC-IDS2017** This dataset will primarily serve as the initial training dataset. It contains a wide variety of realistic network traffic, including benign traffic and several common attack types such as DoS, DDoS, brute force attacks, web attacks, botnets, and Heartbleed. The data features are extracted using the CICFlowMeter tool.
- **Dataset 2: CSE-CIC-IDS2018** This dataset will be used for fine-tuning and final evaluation, particularly to test the transfer learning capability and the detection of novel attacks. It also contains realistic traffic and a broad spectrum of attacks, including many found in CIC-IDS2017, but often with newer variants or under different scenarios. This dataset also utilizes features extracted by CICFlowMeter, ensuring feature compatibility with CIC-IDS2017.

Implementation Guidelines and Expected Delivery

Here is a breakdown of implementation steps for each task:

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Phase 1: Data Acquisition and Preprocessing

- **Data Understanding:** Acquire CIC-IDS2017 and CSE-CIC-IDS2018 datasets. Research and understand the network traffic features (CICFlowMeter) and the cyberattacks within the datasets.
- **Preprocessing Pipeline:**
 - Implement data cleaning (handle missing values).
 - Encode categorical features.
 - Scale numerical features.
 - Ensure feature consistency and compatibility between CIC-IDS2017 and CSE-CIC-IDS2018.
 - Perform feature selection/engineering if necessary to identify meaningful inputs.

Phase 2: Base DNN Training

- **Initial DNN Design:** Select and define an appropriate deep learning architecture for network traffic classification.
- **Implementation:** Implement the chosen DNN architecture using a deep learning framework.
- **Training:** Train the DNN classifier on the preprocessed CIC-IDS2017 dataset to distinguish between normal and known malicious traffic patterns.
- **Evaluation:** Evaluate the performance of the base DNN on a test set from CIC-IDS2017 using relevant metrics.

Phase 3: Transfer Learning

- **Adaptation:** Take the pre-trained DNN from Phase 2. Its architecture, especially the output layers, must be adapted to accommodate new attack types present in CSE-CIC-IDS2018.
- **Fine-Tuning:** Fine-tune the adapted DNN using the preprocessed CSE-CIC-IDS2018 dataset.
- **Evaluation:** Evaluate the fine-tuned model's performance on a test set from CSE-CIC-IDS2018, paying attention to its ability to detect novel attacks.

Phase 4: FLS Design

- **Approach Selection:** Decide what information the FLS will use for its interpretable reasoning. This choice determines whether the FLS interprets raw data characteristics or the DNN's learned representations and decisions. Choose one primary source:
 - **Interpret Original Input Features:** The FLS analyzes selected raw network traffic features after preprocessing. Rules connect feature values (e.g., high packet rate) to fuzzy concepts (e.g., likely DoS).
 - **Interpret DNN Information:** The FLS analyzes data from the trained DNN, such as the final output probabilities/scores. Rules connect DNN state/output (e.g., high DoS probability) to fuzzy concepts.

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- **FLS Design:**

- Define linguistic variables corresponding to the chosen inputs (features or activations).
- Establish appropriate membership functions for each linguistic variable.
- Develop fuzzy rules based on domain knowledge or insights derived from the DNN's behavior.

- **Implementation:** Implement the fuzzy inference engine.

Phase 5: Hybrid System Integration

- **Integration:** Design and implement the architecture to combine the fine-tuned DNN and the FLS into a unified hybrid IDS. This might involve using the FLS to interpret/adjust DNN outputs or combining decisions from both components.
- **Final Evaluation:** Comprehensively evaluate the final hybrid system on the CSE-CIC-IDS2018 test set. Assess:
 - Overall detection accuracy and performance metrics.
 - Effectiveness in detecting novel attack types.
 - The interpretability provided by the fuzzy logic component.

Document the entire process, including data preprocessing, model architectures (DNN and FLS), training/fine-tuning procedures, integration strategy, and evaluation results. Prepare the final project report.